

COMPLETED AND PRELIMINARY FOUNDATIONS



EXPERIMENT-CENTERED AIMS

Measured 4D grain evolution is the scientific target

Local learning is the tool; experimental behavior defines the claims



AIM 1

Establish reliable experimental learning

Spatial reproducibility, baselines, and applicability limits



AIM 2

Validate rollout on unseen experimental partitions

Frozen protocol, training-blind test, and per-family adjudication



AIM 3

Interpretability and physics learning

Validated responses advance toward physical attribution